

Microbial keratitis: uncertain organism and treatment failure

Part 2 | When keratitis does not improve

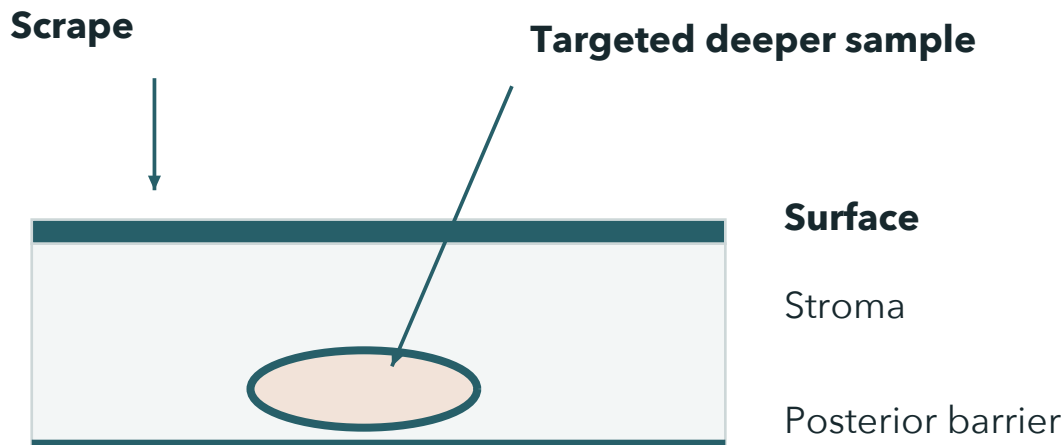
Reassess the trajectory before merely changing the drug.

Track separately	Why it matters
Epithelial defect	Healing, toxicity and exposure
Infiltrate depth and border	Organism burden or wrong diagnosis
Thinning and anterior chamber	Structural threat or intraocular extension
Adherence and delivery	Prescription is not actual exposure

Compare serial drawings or photographs under similar conditions. A smaller epithelial defect with deeper stromal progression is not reassuring. Review contact-lens practices, prior antimicrobial exposure, immunosuppression and the possibility of fungal or amoebic disease. Escalation should address the missing mechanism, not simply intensify an already toxic regimen.

A negative scrape can miss the lesion

Sample the involved compartment, with laboratory coordination.



A superficial negative result may miss a deep focus.

Original section: surface material may not represent a deep stromal focus.

Scraping yields depend on lesion accessibility, pretreatment and processing. Persistent deep disease despite negative superficial samples may justify a targeted corneal biopsy in specialist hands. Confocal microscopy can add structural clues, particularly for difficult organisms, but operator-dependent appearances neither provide susceptibility testing nor reliably replace microbiology.

Recognising a filament is not enough

Morphology directs tests; susceptibility directs treatment.

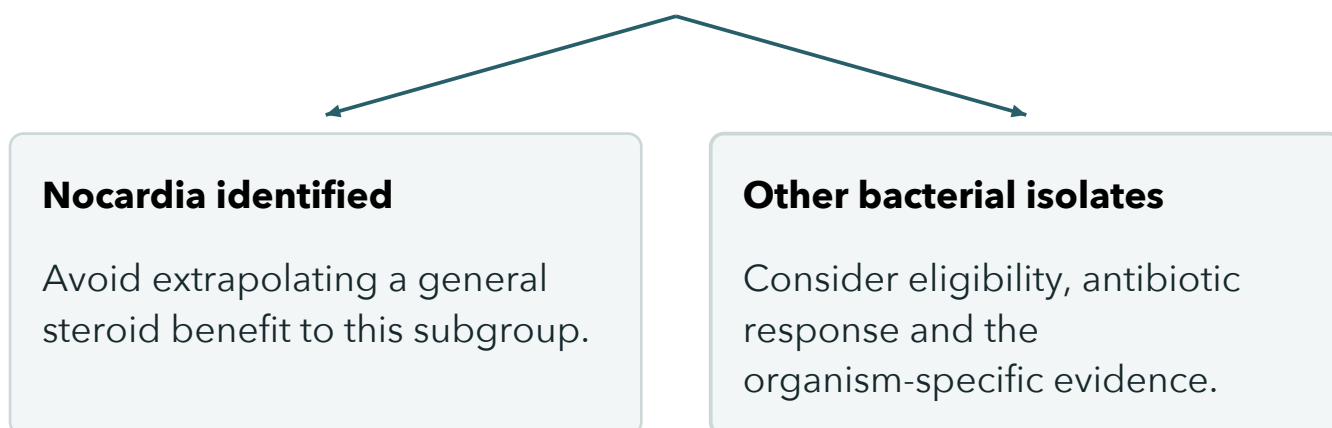
Finding	Interpretation
Beaded branching Gram-positive filaments	Consider Nocardia; request appropriate stains
Partial acid-fastness	Supports Nocardia, not a universal positive result
Filamentous fungal elements	Different organism class and drug pathway
Wreath-like infiltrates	Suggestive pattern, not definitive proof

Tell the laboratory what is suspected. Clinical morphology overlaps, and organisms can be obscured by prior treatment. Amikacin often has a role in Nocardia keratitis, but isolate susceptibility and clinical response matter. Do not extrapolate a bacterial regimen to every branching organism.

The steroid question has an organism interaction

Trial averages do not erase subgroup harm.

Would a bacterial label justify adjunctive steroid?



The Nocardia analysis within SCUT associated adjunctive corticosteroid with a larger residual infiltrate or scar. This is a subgroup finding within a randomized trial, not proof that every bacterial ulcer must receive or avoid steroid. Fungal and amoebic infection require separate reasoning; suppressing inflammation while viable organisms persist can be dangerous.

Route escalation needs an outcome rationale

More systemic exposure is not automatically more corneal benefit.

Question	Decision relevance
Does topical delivery reach the focus?	Depth and susceptibility must be reconsidered
Is there systemic or intraocular spread?	Different treatment indication
Is tissue structurally failing?	Antimicrobial therapy alone may not seal it
What endpoint improved in the trial?	Healing, perforation and vision are distinct

MUTT II did not establish an overall reduction in perforation or therapeutic keratoplasty from adding oral voriconazole to topical therapy for severe filamentous fungal ulcers, while systemic adverse effects increased. This does not exclude targeted systemic treatment for a different indication. Separate antimicrobial control from urgent tectonic rescue.